

Day 4: EC2 Purchasing Models & Alternative Compute

Spot/Dedicated Hosts, AWS Lambda, Lightsail, and Elastic Beanstalk

Module 1: EC2 Pricing Models

AWS provides four main purchasing configurations to optimize costs. Choosing the correct model can reduce billing rates for compute resources significantly.

- **On-Demand Instances:** Pay for compute capacity by the second or hour. No upfront costs or long-term commitments. Ideal for unpredictable, short-term testing environments.
- **Reserved Instances / Savings Plans:** Commit to a baseline level of compute usage (in \$/hr) for a 1 or 3-year term. Reduces cost by up to 72% compared to On-Demand rates.
- **Spot Instances:** Bid on spare AWS server capacity. Save up to 90% off On-Demand rates. However, AWS can reclaim the server with a 2-minute warning. Suitable for batch processes and stateless microservices.
- **Dedicated Hosts:** Physical EC2 servers dedicated fully to a single user. Helps meet strict compliance policies and allows you to use existing host-bound software licenses (BYOL).

Module 2: AWS Compute Alternatives (Serverless & PaaS)

Manually scaling virtual servers is not always required. AWS provides managed abstractions to deploy code without manual server maintenance.

AWS Service	Abstraction Model	Management Overhead	Key Characteristics
AWS Lambda	Serverless (FaaS)	Minimal (AWS manages runtimes)	Event-driven; scales automatically; pay-per-millisecond; 15-minute runtime limit
Elastic Beanstalk	Platform as a Service	Low (AWS handles provisioning)	Upload source code; Beanstalk deploys host, DNS, auto-scalers; you maintain controls
Amazon Lightsail	Virtual Private Server	Medium (Fixed OS configs)	Simplified virtual servers; fixed, low-cost monthly plans; bundle storage/DNS